

Pop-up bubbles

X25 (2025) — English translation

When observing air bubbles floating in a liquid, you will notice that their shape very much depends on the volume of air inside the bubbles and can be either close to spherical or resemble a thin disk. This problem is devoted to the study of the shapes of air bubbles floating in a liquid. When solving the problem, we will accept the following model of the phenomenon: unless otherwise indicated, all moving bodies have a shape slightly different from spherical; the compressibility and viscosity of the liquid can be neglected; the gravitational field in parts A and B can be neglected.

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Part A. Well forgotten old (5.0 points)

Let us consider a ball of radius R moving with speed $\vec{v}_0$ in an infinite volume of incompressible fluid with density ρ and zero viscosity. The ball sets the liquid in motion. Recall that the gravitational field can be neglected. The movement of the fluid is characterized by the quantity μ , called the added mass. The following characteristics of a moving fluid are associated with the added mass: If a ball moves in a fluid with a speed $\vec{v}_0$, then the kinetic energy E_k of the fluid surrounding the ball is determined by the expression:

$$E_k = \frac{\mu v_0^2}{2}.$$

Если шар движется в жидкости со скоростью $\vec{v}_0$, то импульс $\vec{p}$ жидкости, окружающей шар, определяется выражением

$$\vec{p} = \mu \vec{v}_0.$$

Пусть $\vec{r}$ — радиус-вектор, проведенный из центра шара в произвольную точку пространства, а $\vec{v}(\vec{r})$ — скорость соответствующего кусочка жидкости в лабораторной системе отсчёта. При решении задачи вам понадобится теорема Томсона, утверждающая, что если изначально неподвижную несжимаемую жидкость с нулевой вязкостью привести в движение, то циркуляция вектора скорости жидкости по любому замкнутому контуру, не пересекающему поверхности движущихся в жидкости тел, обращается в ноль, что можно записать следующим образом:

$$\oint_l \vec{v} d\vec{l} = 0.$$

В пунктах A1 - A9 будем решать задачу в в системе отсчёта, связанной с шаром. Скорость жидкости в этой системе отсчёта будем обозначать как $\vec{v}_{\text{rel}}$.

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- A1** What is the flow of the relative velocity vector of the fluid through an arbitrary closed surface that does not intersect the surface of the ball? Justify your answer. Note: the flow of the relative fluid velocity vector through a closed surface is the quantity Φ , determined by the expression: **0.30**

$$\Phi = \oint_S \vec{v}_{\text{rel}}(\vec{r}) d\vec{S}.$$

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- A2** What is the normal component of the fluid velocity $v_{\text{oth},n}$ in the frame of reference associated with the ball? **0.10**

- A3** What is the relative velocity of the fluid $\vec{v}_{\infty}$ at an infinite distance from the ball? **0.10**

- A4** Let us consider a superconducting body placed in a uniform magnetic field with induction $\vec{B}_0$. Write an expression for the circulation of the magnetic field $\vec{B}(\vec{r})$ along a contour that does not intersect the surface of the superconductor, and an expression for the flux of the magnetic field through a closed surface that does not intersect the surface of the superconductor. Also indicate the boundary conditions of the magnetic field at the boundary of the body and at an infinite distance from it. **0.40**

- A5** Let us place a superconducting ball of radius R in a uniform magnetic field with induction $\vec{B}_0$. Find the magnetic field induction $\vec{B}(\vec{r})$ in the entire space outside the sphere. Express your answer in terms of $\vec{B}_0$, $\vec{r}$ and R . **0.80**

- A6** Using the results of the previous paragraphs, find the dependence $\vec{v}_{\text{oth}}(\vec{r})$. **0.20**

Since the ball moves in the fluid at a constant speed, in the ball's frame of reference the motion of the fluid is stationary. This allows us to obtain the pressure distribution on the surface of the ball.

- A7** Let us consider the stationary motion of an incompressible fluid with zero viscosity through a thin tube. Let p be the pressure in the liquid, φ the potential energy per unit mass of the liquid, and v the flow velocity in the pipe. Using the laws of conservation of energy and mass, show that for all points of the tube the relation (Bernoulli equation) holds: **0.40**

$$\varphi + \frac{p}{\rho} + \frac{v_{\text{OTH}}^2}{2} = \text{const}$$

Let θ be the angle between the radius vector $\vec{r}$ and the speed $\vec{v}_0$.

- A8** Assuming that the pressure at infinite distance from the ball is equal to p_0 , find the dependence $p(\theta)$ on the surface of the ball. Express your answer in terms of p_0, ρ, v_0, θ . **0.60**

- A9** Calculate the resultant of the pressure forces acting on the surface of the ball. Express your answer in terms of $\vec{v}_0, R, \rho$. **0.40**

Let's return to the laboratory frame of reference. In the laboratory reference frame, the fluid velocity at a point with radius vector $\vec{r}$ is equal to $\vec{v}(\vec{r})$

- A10** Find the distribution of fluid velocity $\vec{v}(\vec{r})$ in the laboratory frame of reference. **0.10**

- A11** What is the kinetic energy of the liquid E_k in the laboratory frame of reference? Express your answer in terms of ρ, R, v_0 . **0.80**

- A12** Find the added mass μ . Express your answer in terms of ρ, R . **0.20**

If you were unable to solve the previous point, you can continue to count $\mu = \frac{4}{3}\pi R^3 \rho$.

- A13** Consider a ball moving with speed $\vec{v}_0$ and acceleration $\vec{a}_0$. Using the expression for the momentum $\vec{p}$ of the fluid, obtain an expression for the force $\vec{F}$ acting on the ball from the fluid. Express your answer in terms of ρ, R and $\vec{a}_0$. **0.30**

Since the bubble is not a solid body, it will be deformed as a result of movement. In the future, consider its deformation to be small. Let us consider the movement of a bubble with a surface tension coefficient σ with a speed v_0 . Consider all expressions obtained in the previous paragraphs to be applicable. The pressure inside the bubble is the same everywhere and is equal to p_{in} .

- A14** Obtain the dependence of the curvature K of the bubble surface on the angle θ . Use $p_{in}, p_0, \rho, \theta$ and v_0 . Note: the curvature K of a surface is the sum of the inverse principal radii of curvature: **0.30**

$$K = \frac{1}{R_1} + \frac{1}{R_2}$$

Она связана с Лапласовым давлением как:

$$\Delta p = \sigma K$$

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Part B. Effect of speed on bubble deformation (2.5 points)

Let us consider an ellipsoid obtained from an ellipse with a semi-major axis a and a semi-minor axis b by rotation about the minor axis. We denote the rotation axis as y . This figure can be obtained by compressing a sphere of radius a along the axis y with a coefficient k , where $k = b/a$. Let us find the curvature of the surface of the resulting figure at an arbitrary point A' , which before compression corresponded to the point A and the angle θ , measured from the axis y .

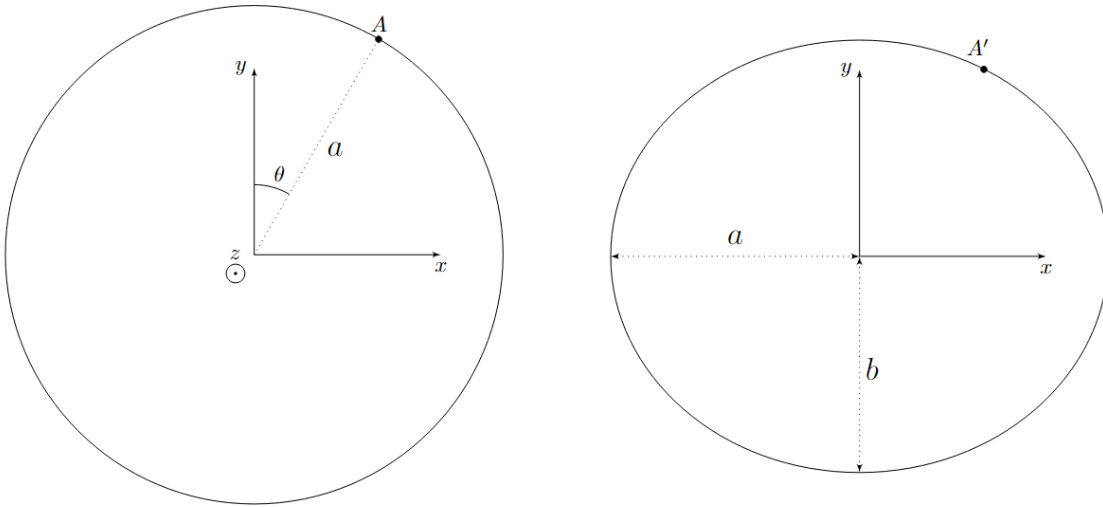


Fig. 1.

It can be shown that in the (x, y) plane the radius of curvature of the ellipsoid is equal to:

$$R_1(\theta) = \frac{a}{k} \cdot (\cos^2 \theta + k^2 \sin^2 \theta)^{\frac{3}{2}},$$

а в плоскости, содержащей нормаль к эллипсоиду в точке A' и перпендикулярной плоскости (x, y) , радиус кривизны эллипсоида равен:

$$R_2(\theta) = \frac{a}{k} \sqrt{\cos^2 \theta + k^2 \sin^2 \theta}.$$

Так как $K(\theta)$ в части А было выведено в предположении, что капля является сферой, то отклонения от сферы должны быть маленькими. Это означает, что $k = 1 - \varepsilon$, где $\varepsilon \ll 1$. Также считайте, что $a \approx R$, где R is the radius of the bubble we used in Part A.

B1 Taking into account the smallness of ε , the quantities R_1 and R_2 can be represented in the form: **0.60**

$$R_1(\theta) \approx R(1 + A_1\varepsilon - B_1\varepsilon \sin^2 \theta) \quad R_2 \approx R(1 + A_2\varepsilon - B_2\varepsilon \sin^2 \theta).$$

Find B_1 and B_2 . In what follows, the terms $A_1\varepsilon$ and $A_2\varepsilon$ can be neglected. If you were unable to solve this point, then you can further assume that $B_1 = B_2 = 1$. This does not lead to the loss of subsequent points.

B2 Find the dependence of the curvature of the ellipsoid $K(\theta)$ in first order in ε . Express your answer in terms of R , θ . **0.20**

B3 What is the value of ε for the bubble discussed in A14? Express your answer in terms of the bubble speed v_0 , bubble radius R , surface tension coefficient σ and liquid density ρ . **0.60**

B4 What is the eccentricity e of the bubble discussed in A14? Express your answer in terms of the bubble speed v_0 , bubble radius R , surface tension coefficient σ and liquid density ρ . **0.30**

B5 A spherical bubble with an initial diameter $d = 2\text{mm}$ rises in water with density $\rho = 10^3\text{kg/c}$ and surface tension $\sigma = 72 \cdot 10^{-3}\text{ H/M}$. After taking the necessary measurements, calculate the speed of the bubble in the last three photographs. Also calculate the acceleration of the bubble a . **0.80**

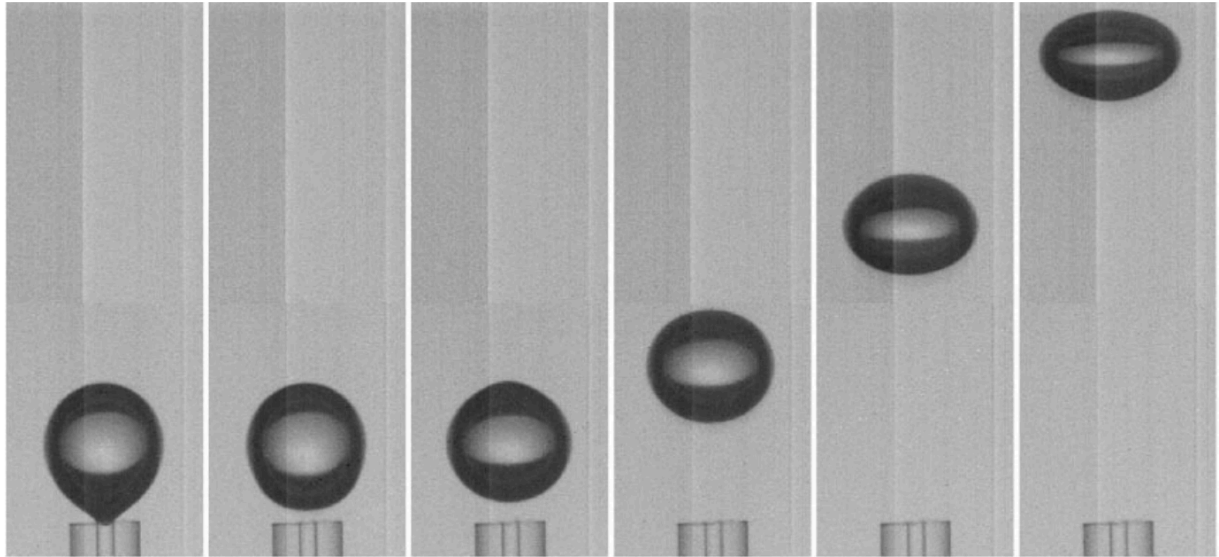


Fig. 2.

Part C. Dynamics of Bubble Motion (1.5 points)

In real life, the free fall acceleration g will create a vertical pressure gradient in the vessel, as a result of which the Archimedes force F_A will appear, accelerating the bubble upward. Initially, the bubble is spherical, but during acceleration, according to the theory from the previous paragraph, it begins to flatten. Newton's second law for a nearly spherical bubble would look like this:

$$m\vec{a} = m\vec{g} + \vec{F}_A + \vec{F}_\mu + \vec{F}_{\text{TP}},$$

где m - масса пузыря, которой можно пренебречь, $\vec{F}_A$ - сила Архимеда, $\vec{F}_\mu$ - сила, действующая со стороны воды на пузырь, связанная с присоединённой массой воды, $\vec{F}_{tr}$ - сила трения, возникающая из-за эффектов турбулентности. В нашей модели $\vec{F}_{tr}$ хорошо описывается формулой $\vec{F}_{tr} = -\pi R^2 \rho v \vec{v}$, где $\vec{v}$ is the bubble speed. The speed of the bubble is initially zero.

C1	What is its acceleration a at the initial moment of time? Express your answer in terms of g .	0.60
C2	Find the steady speed of the bubble $v_{\max}$. What is the value of ε at steady speed? Express your answers in terms of R , g , ρ and σ .	0.30
C3	Calculate the numerical values of $v_{\max}$ and ε for $\rho = 10^3 \text{ кг/м}^3$, $\sigma = 72 \cdot 10^{-3} \text{ Н/м}$, $g = 10 \text{ м/с}^2$ and $R = 1 \text{ мм}$. Is the selected model applicable to the specified radius?	0.60

Part D. Cool Jellyfish Bubble (2.0 points)

In large-volume bubbles, the force of surface tension is weaker than in small ones, so their shape when they reach a constant speed is far from spherical.

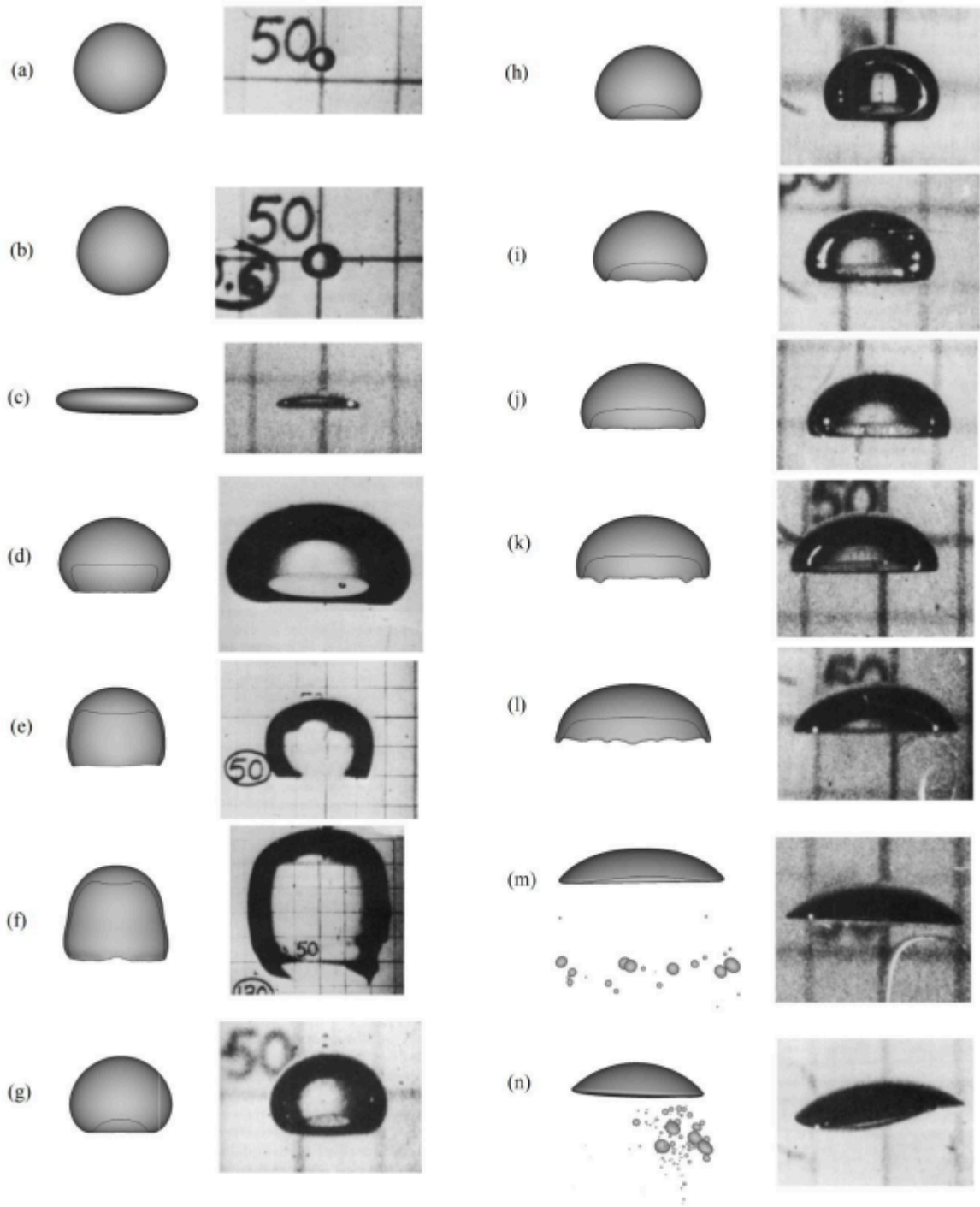


Fig. 3.

In general, the shape of the bubbles shown in the figure above is simulated using a computer. However, in the previous parts we were able to study a shape close to spherical: these are the bubbles *a* and *b*. In this part we will study the model applicable to bubbles *m* and *n*. Let us describe the bubble as a segment of a sphere with base radius r . The radius of curvature of the bubble, that is, the radius of the sphere, is constant and equal to R . Photographs *m* and *n* show that the bubbles are almost flat, which means we can assume that $r \ll R$. We will characterize a point on the bubble surface by a small angle θ , measured from the center of curvature. The air pressure inside the bubble is p_{in} .

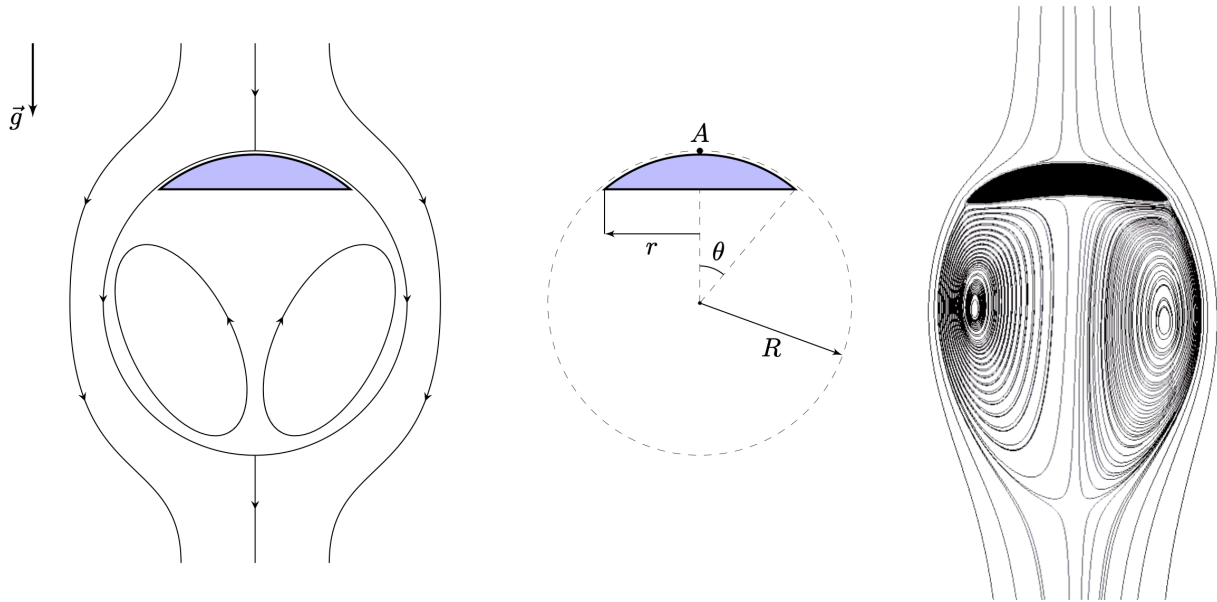


Fig. 4.

D1 Prove that the water pressure at points on the surface of the bubble is constant.

0.20

The water flow lines are schematically indicated in the figure. We will assume that the distribution of fluid velocities $\vec{v}_{\text{OTH}}(\vec{r})$ over the bubble is exactly the same as in the case of motion of a solid ball of radius R with speed v_0 . Gravity acceleration g .

Let the bubble float in a deep pond. The pressure on the surface of the reservoir is equal to atmospheric pressure p_0 . The distance from the surface of the reservoir to the point A is $h \gg R$.

D2 Find the pressure relationship above the bubble surface $p(\theta)$ using ρ , v_0 , g , R , θ and h .

1.00

D3 Derive an expression for the steady-state velocity of the bubble v_0 . Calculate the value for $R = 30 \text{ cm}$ and $g = 10 \text{ m/c}^2$.

0.80

Source: <https://pho.rs/p/4196>